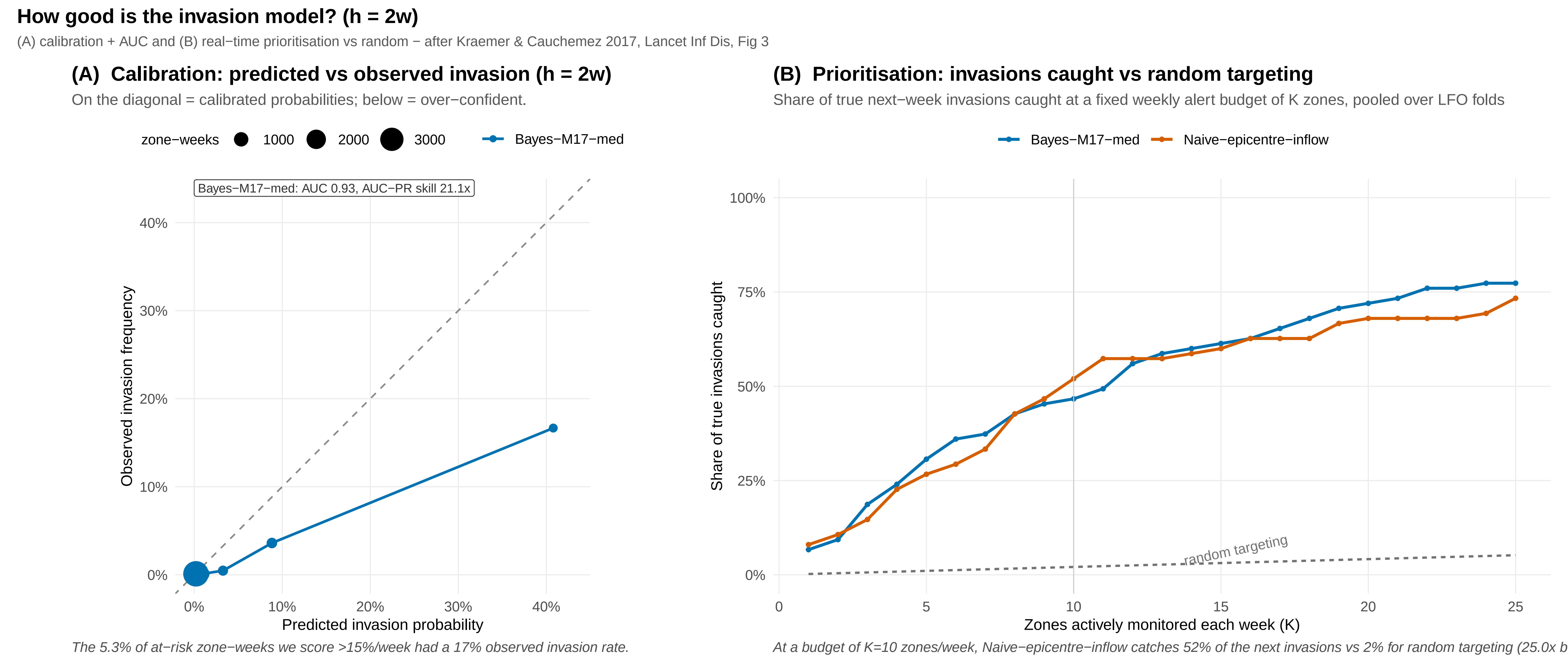
How good is the invasion model? (h = 2w)
(A) calibration + AUC and (B) real−time prioritisation vs random − after Kraemer & Cauchemez 2017, Lancet Inf Dis, Fig 3
(A) Calibration: predicted vs observed invasion (h = 2w)
On the diagonal = calibrated probabilities; below = over−confident.
zone−weeks
1000
2000
3000
Bayes−M17−med
Bayes−M17−med: AUC 0.93, AUC−PR skill 21.1x
Observed invasion frequency
40%
30%
20%
10%
0%
0%
10%
20%
30%
40%
Predicted invasion probability
The 5.3% of at−risk zone−weeks we score >15%/week had a 17% observed invasion rate.
(B) Prioritisation: invasions caught vs random targeting
Share of true next−week invasions caught at a fixed weekly alert budget of K zones, pooled over LFO folds
Bayes−M17−med
Naive−epicentre−inflow
Share of true invasions caught
100%
75%
50%
25%
0%
random targeting
0
5
10
15
20
25
Zones actively monitored each week (K)
At a budget of K=10 zones/week, Naive−epicentre−inflow catches 52% of the next invasions vs 2% for random targeting (25.0x b